

First name:

Last name:

Student ID:

Name and Student ID must be filled in for this page to be graded.

Problem 1. *Use the exactness criteria to check if the following vector field is path independent in a simply connected domain:*

$$\vec{F} = x^2y \, \mathbf{i} - 4xy^2 \, \mathbf{j} + 8z^2x \, \mathbf{k}.$$

[5 marks]

Solution.

□

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Problem 2. Find $\oint_C \vec{F} \cdot d\vec{r}$ for the field $\vec{F} = 6y^2 \mathbf{i} + 2x - 2y^4 \mathbf{j}$, where C is the counter-clockwise boundary of the square region given by

$$R = \{(x, y) : -2 \leq x \leq 2, -2 \leq y \leq 2\}.$$

Hint: There is no restriction on how to compute the result, so any valid method will be accepted.

[5 marks]

Solution.

□